In Partnership With

Clinical Squared's response to CMS-0042-NC.

Request for Information; Health Technology Ecosystem

Prepared specifically for

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Clinical Squared, Inc., in collaboration with Visual Connections and BurstIQ, is pleased to submit our response to the **Health Technology Ecosystem RFI**. For clarity, we have included only the questions we've addressed and grayed out those not answered. We are excited to support CMS in advancing a scalable, impactful Health Technology Ecosystem and welcome your feedback.

B. Patients and Caregivers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve patients and caregivers. While we certainly want patients and caregivers to answer questions in this section (and in other sections) from the patient/caregiver point of view, we also invite all stakeholders to provide their viewpoints on the patient/caregiver workflows.

1. Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take? An application that lets the user connect their own health data repositories to public and private datasets and helps create understandable analytic insights between the personal data and the connected data.

a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products? Intermediate on their behalf with healthcare service providers and payers, assist with research, selection, and scheduling of health services/providers, and help to ensure their data is being used appropriately and with consent by the right stakeholders.

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future. See response in question **a.** above

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)? No. Clinical Squared is developing a platform that unifies health information for a patient that is scattered across multiple EMR's, patient portals, and caregiver practice datasets.

a. If so, what are some examples of benefits it has provided? N/A

b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information? Our workflows feature custom API connections that are compatible with major EMR vendors (Epic, Cerner Health, and Meditech), support bulk uploads of data, and provide capability for ad hoc/individual document uploads.

c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)? Images from DICOM/PACS systems tend to be very large and are

typically not available to the patient without making a special request of the health information management (HIM) department of the practice facility. In some cases, full lab reports, doctor's notes, or unedited diagnosis data is not available for fear it might be misinterpreted by a lay person.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers? We are architecting and developing a health data management app we feel would be useful and intuitive to Medicare beneficiaries and are happy to discuss in detail with CMS leadership at their convenience.

PC-4. What features are missing from apps you use or that you are aware of today? Integration with multiple, different healthcare practice EMR systems, scheduling, billing, consent, and data review features on the same platform, and apps that have an intuitive user interface that is easily accessible from a mobile device.

a. What apps should exist but do not yet? Why do you believe they do not exist yet? An app that has true interoperability between disparate EMR vendors, payers, and health care practices that can capture data created about a patient and include contextual data from the patient themselves. This kind of app can be very costly to build because of the variable data types and sources, and the lack of cooperation between different data custodians/software vendors in supporting a truly interoperable application that does not originate from them (see above answers as well).

b. What set of workflows do you believe CMS is uniquely positioned to offer? Workflows around claims data, physician and healthcare service provider data, and public health trend data based on reported outcomes and services utilized.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products? Measure the use of innovative technology tools and support high performing products publicly so that patients and caregivers build trust and the security and accuracy of digital health products that improve the patient and the physicians experience in healthcare service provision.

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)? If CMS were to suggest high performing, accurate, secure digital health products to the public, we think it would be meaningful for CMS to have a review process that includes seeking inputs from the health IT industry (including vendors and software developers as well as patients and providers who use these tools). We also think that tying greater physician reimbursement to the proven, effective use of digital health products will drive practices to demonstrate the efficacy of different apps and platforms.

b. What changes would enable timely access to high quality CMS and provider generated data on patients? Increase the frequency with which high quality data is reviewed and published by CMS. Additionally, partnering with private sector solution developers and software vendors that are producing high quality data that is generated by providers and CMS will relieve the burden on agency staff in reviewing and publishing quality data with its workforce.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services? Ease of use is critical. Digital health products that have an intuitive user interface and provide data that is Accurate and easy for the user to understand are always more successful.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so? Product and market analysis for digital health tools is certainly available from various sectors of the public. Creating an RFI for the health IT industry to respond with analysis seems like a good start.

2. Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both? We have seen the utility in publicly available data from trusted resources such as CMS, NIH, CDC, and the FDA. We are strong proponents of creating solutions for consumers that foster access to this public data in a way that connects to an individual's personal context and informs their health journey.

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)? Data housed in proprietary systems such as electronic medical record (EMR) platforms, clinical research repositories, payor databases, and other commonly commoditized sources are particularly difficult for patients and caregivers to gain access. While an individual may have options of access to their own data from these sources, access to empirical or general data about similar experiences, episodes and plans of care, and the like are often unavailable.

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why? We know patients and caregivers benefit from anecdotal information from other patients and caregivers undergoing similar treatments, providing similar healthcare services under similar circumstances and so on. The popularity of patient support groups, professional caregiver cohorts, and other such entities demonstrate that people like to ask their (relevant) peers about their experiences navigating similar situations in healthcare service delivery.

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence? Opportunities include the availability of technology that can ensure secure, auditable exchanges of sensitive data between patients, their caregivers, insurance companies and other stakeholders in a way which allows them to maintain control of their own data and consent. We are deeply involved in building systems that utilize blockchain technology for digital ID creation, securing digital documents (assets), and the facilitation of these kinds of transactions.

The availability and accessibility of AI tools that can parse enormous data sets quickly and with a high degree of accuracy provide additional opportunities to create tools that make data more consumable and understandable, especially to the general population. We have developed several open-source tools that help condense, simplify, and explain healthcare-centric data using AI algorithms of this sort.

Both examples of technology above can be leveraged to interoperate and integrate in legacy systems so there is no need to incur extreme expense to “rip and replace” in-place systems with new ones.

Some challenges we see around improving accessibility, interoperability, and integration of clinical data from different sources lie in the siloed state of data. We believe that governmental support of, and sometimes mandate for, an increase in accessibility and transparency of data for the benefit of consumers and in the furtherance of public health and safety could increase the availability of valuable data, and in some cases, diminish costs.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers? We think the inclusion of physician demographic and practice information (like that in NPPES) can be particularly useful for patients choosing an appropriate caretaker for their health care scenario.

a. What difficulties are there in accessing or utilizing these data sources today? In our experience, accessing data sources that are available through the Integrated Data Repository (IDR) requires CMS to approve and enable that access for individuals needing or requesting it. While this avenue of access allows for more seamless integration of data from multiple CMS curated data sets, the complexity of this provisioning process can be inefficient. Conversely, CMS data sources that are publicly available are siloed and disconnected in such a way that it is difficult for the public to use them, either by cross-referencing related data or applying their own data context, in a meaningful way.
b. What suggestions do you have to improve the Blue Button 2.0 API experience? We think that expanding the data sources available in the API (as mentioned above) would be a great improvement. We also suggest enhancing developer documentation about the API architecture. Publishing clear, detailed guidance would be useful, especially for developer groups that are newly coding integration for Blue Button 2.0 API
c. Is there non-CMS data that should be included in the API? We believe there is tremendous value in data from FDA (drug and medical device information), NIH (clinical trials data) and CDC (public health and safety information). Additionally, tying the API into private sector data such as Healthgrades and Yelp could lend new options of utility for consumers.

PC-10. How is the Trusted Exchange Framework and Common Agreement™ (TEFCA™) currently helping to advance patient access to health information in the real world?

a. Please provide specific examples.

b. What changes would you suggest?

c. What use cases could have a significant impact if implemented through TEFCA?

d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?

e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?

f. Are there redundant standards, protocols, or channels that should be consolidated?

g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world? The interchange of both routine and critical patient data across various medical systems has the potential to dramatically impact patient outcomes, however patient access tends to lag behind clinician access. We believe the primary reason behind this is the security infrastructure necessary to ensure that the access is protected and the actual patient can be sufficiently verified

a. How valuable, available, and accurate do you find the data they share to be? Historically the accuracy of the demographic and personal information was a main issue, the clinical info was not. This ultimately led to clinicians not relying on the HIE for accurate info.
b. What changes would you suggest? Data normalization is critical. Unfortunately, that needs to start at each facility's EMR solution. Accurate and consistent name (first, last, middle initial and last 4 of SSN or other universal unique identifier) would be a good start.
c. Are there particular examples of high-performing HIE models that you believe should be propagated across markets? The model propagated by the HIE in Maryland called CRISP (Chesapeake Regional Information System for our Patients) has been effective and successful enough to have expanded their services into 5 other states through shared services partnerships. Their example should certainly be considered.

d. What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both? Immediacy of information is critical, weekly or monthly batch uploads just don't help the patients. And again, data normalization would be key to this issue.

a. Examples may include the following:

(1) Binding cost estimates for pre-defined periods. Agreed. Maryland's rate commission helps to address this issue.

(2) Viewing provider schedule availability. Agreed, but again, this relies on normalization of each facility's EMR solution.

(3) Using third-party apps for appointment management. Far too often third-party apps are too focused on one or two solutions to a problem: calendar scheduling and cancellations. Third party apps that consider other necessary elements like EMR integration, real time alerts to changes, and the other features can be useful, especially if they can integrate across multiple systems.

(4) Accessing patient-facing quality metrics. We agree that this is a valuable data set, however, facilities are all over the place tracking this information. Perhaps an operational agreement can help narrow the gap and provide a benefit for the actual patient.

(5) Finding the right provider for specific healthcare needs. Far too often the facilities don't maintain this information on a timely basis, as physician rosters are notoriously out of date for most facilities, payer networks, and the like. Patient-facing search tools claim providers are accepting new patients, but when the patient reaches out to the provider directly, they are not accepting new patients.

b. What use cases are possible today? Patient and provider demographics, and clinical outcomes

c. What should be possible in the near future? If stringent requirements are imposed, the data can be much more valuable to both clinician and patient.

d. What would be very valuable but may be very hard to achieve? The aforementioned data normalization. If all participating systems can provide consistent accurate data, HIEs can play a much larger role in positive and informative patient experiences.

3. Information Blocking and Digital Identity

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange? Determining which patient and caregiver submissions have merit will be challenging and may require resource-intensive manual review. It may be more effective to enforce at the healthcare provider (HCP) level – make claims subject to denial if it is found that the HCP did not use reasonable efforts to obtain information, or penalties (even civil action) if an HCP withheld information.

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):

a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials? A huge challenge lies in lack of trust in government or corporate use of sensitive personal data. This can be mitigated by using solutions like ***Burst IQ's LifeGraph*** in which the patient/caregiver owns and controls their data/identity. Low technology literacy in some populations also proves to be extremely challenging; tech illiteracy and clinical complexity are directly correlated, further exacerbating the issue since these are the populations that use the most services.

b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used? Better care, less duplication of care, increased care coordination, lower costs if digital IDs are used to enable data sharing between HCPs, reduction of the likelihood of identity or PHI theft.

c. What are the potential downsides? Potentially inequitable care for those who can't or don't adopt digital identity solutions. Also, if digital ID credentials are implemented in a centralized manner, the risk of a large-scale breach is significant. This can be mitigated by using a distributed data ecosystem like ***LifeGraph***, which allows patients, providers, and healthcare organizations to own and control their data. This decentralizes data access and control, so breaches can't access as much data.

d. How would encouraging the use of CSPs improve access to health information? By improving the security and authentication that is a critical foundation for all data exchange systems.

e. What role should CMS/payers, providers, and app developers have in driving adoption?

- **CMS/payers:** see 'f' below.
- **Providers:** prioritize EMRs, scheduling systems, workforce solutions, and other systems that leverage digital identities
- **App developers:** make digital identity solutions user-friendly and adopt standards-based solutions.

f. How can CMS encourage patients to get digital identity credentials? Automatically issue a digital identity for all Medicare recipients and require Medicaid agencies to provide a timeline for digital identity adoption. Commercial payers will follow suit.

C. Providers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve providers. While we certainly want providers to answer questions in this section (and in other sections) from the provider point of view, we also invite all stakeholders to provide their viewpoints on the provider workflows as appropriate.

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients? Provide comprehensive support to providers in learning how to leverage approved digital health products and incentivize them to adopt them in a measurable way. Eventually, mandate the use of those digital health products that show a demonstrable improvement in patient experience and patient outcomes.

a. What are the current obstacles? Expense of these novel digital health products, their ability to integrate with legacy systems

b. What information should providers share with patients when using digital products in the provision of their care? Full disclosure of how the products are used by the providers, how the products benefit patient care both from the provider and the patient perspective, and support resources and documentation that is readily consumable and understandable by their patient populations.

c. What responsibilities do providers have when recommending use of a digital product by a patient? Physicians must be very attuned to the utility of the digital products they would recommend to a patient **from the patient's perspective**. Physicians must also understand the risks their patients may assume in using these digital products. We also believe that if a physician recommends the use of a digital product that has an adverse effect, unintended consequences, or causes actual damage in the patient's experience, they should be prepared to assist in mitigating the damage, at least indirectly.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated? Cost is a large obstacle. Effective development, deployment, and utilization of best-in-class innovative applications tend to be very expensive.

Clinicians/practices are also risk-averse and tend to adopt innovation very slowly. Incentivizing adoption of innovation through direct funding of its development and integration could be very motivating in this regard.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

a. Current challenges in accessing different data formats.

b. Impact on patient care quality.

c. Technical barriers to full data accessibility.

d. Cost or privacy implications of making all data formats interoperable.

e. Priority level compared to other interoperability needs.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

2. Data Exchange

PR-5. Which of the following FHIR APIs and capabilities do you already support or utilize in your provider organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints:

a. Patient Access API.

b. Standardized API for Patient and Population Services.

c. Provider Directory API.

d. Provider Access API.

e. Payer-to-Payer API.

f. Prior Authorization API.

g. Bulk FHIR—Do you support Group ID-based access filtering for population-specific queries?

h. SMART on FHIR—Do you support both EHR-launched and standalone app access? What does the process for application deployment entail?

i. CDS Hooks (for clinical decision support integrations).

PR-6. Is TEFCA currently helping to advance provider access to health information?

a. Please provide specific examples.

b. What changes would you suggest?

c. What other options are available outside of TEFCA?

d. Are there redundant standards, protocols or channels or both that could be consolidated?

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

a. What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?

b. In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?

c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

3. Digital Identity

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship? Mandate it. This will force EMRs to incorporate it into their solutions. Providers will only adopt if the EMRs provide it as part of their solution.

a. What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider? See above. Also, this requires patient/consumer adoption, which will be driven by states (i.e., Real ID)

b. How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health? Allow patients to opt-in to personalization of recommended services, potentially in exchange for copay credits or other financial incentives. This is a familiar paradigm for patients since they are used to personalization in their online retail experiences. ***LifeGraph*** can be

used to power patient-centric service marketplaces where patients can access services that are recommended for them and personalized to them. Data is only shared with the service providers if the patient opts in. Recommendations are based on each person but also based on what has worked/not worked for millions of other similar people. This improves engagement, adherence, and clinical outcomes.

PR-10. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs):

a. What are the challenges and benefits for providers**?** **Challenges:** lack of understanding of how digital identity credentials work impacts adoption. Many providers don't understand if/how digital identity credentials meet their compliance requirements, so they deem it "safer" to stick with legacy manual processes, even though digital identities are more secure, auditable, and trustworthy than manual credentials. **Benefits:** reducing fraud, waste, and abuse, streamlines the credentialing process to get providers working faster.

b. How would requiring their use improve access to health information? Better care, less duplication of care, increased care coordination, lower costs if digital IDs are used to enable data sharing between HCPs. ***LifeGraph*** can serve as the secure data ecosystem that enables patients, providers, and payers to access and share data while maintaining ownership and control of their data. Serving as a national HIE and provider credential network. Data is only accessible to providers who are properly credentialed. Governance and regulatory compliance, including patient consent, are automatically enforced. All transactions are auditable.

c. What are the potential downsides? **Short term:** potentially lengthy adoption and learning curve. **Long term:** potentially inequitable care for patients who can't or don't adopt digital identity solutions**.** This could mean poor *HEDIS* and *eCQM* ratings for providers that serve low-tech-literacy populations.

d. What impact would mandatory credentials have on a nationwide provider directory? Mandatory credentials would be a driving force to make a comprehensive provider directory more useful. Digital credentials would transform a simple directory into a valuable credential repository that drives next-gen credentialing services, healthcare workforce solutions, and other operations software. **LifeGraph** can serve as the secure data ecosystem that allows providers to own, maintain, update, and share their verified identity and credentials. Credentialing services, EMRs, workforce management platforms, and other products and services would be able to access the CMS data ecosystem via API and securely access up-to-date provider data on a real-time basis.

e. How could digital identity implementation improve provider data flow? By providing better authentication/transparency/accountability for provider-to-provider data exchange. Additionally, having a provider digital identity linked to provider activity (encounters, claims, etc.) allows for better understanding of how providers are interacting with each other and with patients. This provides better analytics opportunities for FWA analysis. Because **LifeGraph** can manage individual data (e.g., provider digital identities) and operations data, it can be used to automate many processes that require identity verification and "need to know" standing, all based on unified knowledge graphs of providers, patients, encounters, claims, operational data, and more.

f. Would combining FHIR addresses and identity improve data flow? Yes. Coupling these together would allow for end-to-end automation, per '**e.**' above.

PR-11. How could members of trust communities [3] (for example, QHINs, participants and subparticipants in TEFCA, which requires Identity Assurance Level 2 (IAL2) via Credential Service Providers (CSPs)) better

support the goals of reduced provider and patient burden while also enhancing identity management and security? These entities mentioned already serve as data brokers, so they are in a great position to serve as identity brokers, tying data sharing to identity verification.

4. Information Blocking

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

PR-13. For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?

PR-14. How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

D. Payers

PA-1. What policy or technical limitations do you see in TEFCA? What changes would you suggest to address those limitations? To what degree do you expect these limitations to hinder participation in TEFCA?

PA-2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?

PA-3. How can CMS encourage payers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins?

PA-4. What would be the value to payers of a nationwide provider directory that included FHIR end points and used digital identity credentials?

PA-5. What are ways payers can help with simplifying clinical quality data responsibilities of providers?

a. How interested are payers and providers in EHR technology advances that enable bulk extraction of clinical quality data from EHRs to payers to allow them to do the calculations instead of the provider-side technology?

b. In what ways can the interoperability and quality reporting responsibilities of providers to both CMS and other payers be consolidated so investments can be dually purposed? Are there technologies payers might leverage that would support access to real time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

PA-7. How can CMS encourage payers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

E. Technology Vendors, Data Providers, and Networks

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve technology vendors, data providers, and networks. While we certainly want technology vendors, data providers, and networks to answer questions in this section (and in other sections) from their point of view, we also invite all stakeholders to provide their viewpoints on the technology vendor, data provider, and network use cases as appropriate.

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers? Continue to embrace development of solutions using open-source technology and increase adoption of innovative tools from small innovators/developers with limited deployment, low-cost pilots that prove their safety and efficacy. Also, developer interest in building digital health products for Medicare beneficiaries and caregivers will certainly increase as CMS continues to engage with the developer community, not just the largest healthcare software vendors on the market.

TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs? Patient experience data, quality measures, drug and medical device information, and an accurate physician directory that offers practice information (languages spoken, insurance accepted, etc.)

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API? Data that the patient can provide regarding their health experiences or requirements, or that a provider can provide without

c. What obstacles prevent accessing these data sources today? Availability of secure, user-friendly data interfaces where patients can share their data safely while maintaining control and ownership.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers? Clinical Squared has created custom API's that tie into public data sets from CMS, FDA, NIH, and can be customized to integrate in virtually any healthcare ecosystem. We invite CMS to discuss this in detail with us at its convenience.

2. Digital Identity

TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits? Significant challenges lie in lack of trust in government or corporate use of sensitive personal data, and low technology literacy in some populations which can hamper adoption (*see answers in section* **PC-14 a.**) Some benefits of digital identity implementation include better care, less duplication of care, increased care coordination, lower costs if digital IDs are used to enable data sharing between HCPs, reduction of the likelihood of identity or PHI theft (*see answers in section* **PC-14 b**).

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange? Cybersecurity will be enhanced by

mandating the use of digital ID credentials and standardizing the authentication methodology (as mentioned in question **c.** below). Data exchange will be more highly auditable, and transactions can readily be attributed to specific performers within the data exchange ecosystem.

c. What impact would mandatory use of the OpenID Connect identity protocol have? Mandating the use of this protocol will standardize the way technology developers and vendors build authentication functionality into their tools & applications and will simplify the integration of different external tools into existing systems. It will also ensure that the authentication method used by apps for health data exchanges achieve a benchmark of quality of security in order to be implemented.

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

TD-6. What unique interoperability functions does TEFCA perform?

a. What existing alternatives should be considered?

b. Are there redundant standards, protocols or channels or both that should be consolidated?

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

a. Does it contain the full extent of data elements you need?

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

b. What would be the drawbacks?

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or "EHI export" for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

a. Should this capability be revised to specify standardized API requirements for EHI export?

b. Are there specific workflow aspects that could be improved?

c. Should CMS consider policy changes to support this capability's use?

4. Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

a. What are the primary obstacles to this?

b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

TD-14. Regarding networks' use of FHIR APIs:

a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

TD-15. Regarding bulk FHIR APIs:

a. How would increased use of bulk FHIR improve use cases and data flow?

b. What are the potential disadvantages of their use?

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

a. Do current rules encourage scalable network participation?

b. What changes would improve alignment (for example, API unification, reciprocal access)?

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

5. Compliance

TD-18. Information blocking:

a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response?

b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?

c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

TD-19. Regarding price transparency implementation:

a. What are current shortcomings in content, format, delivery, and timeliness?

b. Which workflows would benefit most from functional price transparency?

c. What improvements would be most valuable for patients, providers, or payers, including CMS?

d. What would further motivate solution development?

F. Value-Based Care Organizations

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve value-based care organizations. While we certainly want value-based care organizations to answer questions in this section (and in other sections) from the value-based care provider point of view, we also invite all stakeholders to provide their viewpoints on the value-based care workflows as appropriate.

1. Digital Health Adoption

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs? Remove the restrictions that insurance plans use to limit patient choices and movement. Several incentives can be implemented:

- **Tie reimbursement and shared savings more directly to the use of digital tools**,
- **Offer additional bonuses** or shared savings for ACOs that achieve high rates of patient engagement with digital tools or measurable improvements in quality metrics attributable to digital interventions,
- Provide ACOs with **enhanced access to real-time data, analytics, and interoperability tools**, making it easier to integrate digital health solutions into care delivery and coordination,
- **Reward ACOs that successfully engage patients through digital means** (e.g., mobile apps, remote monitoring, care navigation platforms), as consumer engagement is increasingly recognized as a key driver of value-based care success.

Obstacles preventing broader digital product adoption include

- The inability for solutions to fit into existing clinical and administrative workflows,
- Limited interoperability between digital tools, EHRs, and other health IT systems,
- If digital tools are not user-friendly or do not demonstrate clear value, utilization rates remain low.

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements? Mandating and incentivizing the use of platforms and software solutions that support advanced analytics, the use of responsible and explainable AI, and population health tools will certainly foster their integration into APM requirements. Making population health analytics and risk stratification core APM requirements will lead to using these technologies in identifying high risk patients, targeting interventions, and efficiently allocating resources. Tying provider reimbursement to the successful use of patient engagement tools like patient portals & digital communication and including metrics that demonstrate these successes as part of APM performance measures can foster the adoption.

VB-3. What are essential health IT capabilities for value-based care arrangements? We believe all examples cited below are essential health IT capabilities for a value-based care practice to positively impact patient outcomes, increase efficiency of care, and lower costs including *care planning*, *patient event notification*, *data extraction/normalization*, *quality/performance measurement*, *access to claims data*, *attribution and patient ID*

matching, and *remote device interoperability*. We would like to add the critical need for health IT to include capturing patient experiences in structured, semi-structured, and unstructured data.

a. Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools.

b. What other health IT capabilities have proven valuable to succeeding in value-based care arrangements? Telemedicine and virtual care, patient safety and incident reporting systems, and clinical decision support (CDS) tools have all been very useful in successful value-based care arrangements.

VB-4. What are the essential data types needed for successful participation in value-based care arrangements? *Structured data* (e.g., numerical, coded, and tabular formats), *semi-structured data* (as in clinical notes, images, and reports), and *unstructured data* (like free-text narratives, social media, and patient-generated content) cover the broad formats from the original sources that data comes from, and so, all are essential. That said, essential data types include:

- Electronic Health Records (EHRs)
- Claims Data
- Social Determinants of Health (SDoH) Data
- Patient Demographics
- Clinical Risk Scores and Health Risk Assessments
- Admission, Discharge, and Transfer (ADT) Data
- Quality Performance Metrics
- Patient Engagement and Experience Data
- Attribution and Patient ID Matching Data

2. Compliance and Certification

VB-5. In your experience, how do current certification criteria and standards incorporated into the ONC Health IT Certification Program support value-based care delivery?

VB-6. What specific health information technology capabilities that could benefit APMs are not currently addressed by existing certification criteria and standards that should be included under the ONC Health IT Certification Program?

VB-7. How can technology requirements for APMs, established through CEHRT or other pathways, reduce complexity while preserving necessary flexibility?

VB-8. How can other HHS policies supplement CEHRT requirements to better optimize the use of digital health products in APMs? As an example, requirements under the Conditions of Participation for hospitals (42 CFR 482.24(d)) require hospitals to transmit electronic patient event notifications to community providers. What barriers are in place preventing APM participants from receiving the same notifications?

VB-9. What technology requirements should be different for APM organizations when comparing to non-APM organizations (for example, quality reporting, and interoperability)? APM and non-APM organizations should

have identical technology and reporting requirements. Doing otherwise would support a two-tier healthcare delivery system which would be unbalanced, cumbersome to administer, and unnecessarily more expensive.

VB-10. In the Calendar Year (CY) 2024 Physician Fee Schedule final rule (88 FR 79413), CMS established that CEHRT requirements for Advanced APMs beyond those in the "Base EHR" definition should be flexible based on what is applicable to the APM that year based on the area of clinical practice. What certification criteria should CMS identify under this flexibility for specific Advanced APMs, or for Advanced APMs in general? Are there specific flexibilities or alternatives to consider for smaller or resource-constrained (such as rural) providers in meeting CEHRT requirements without compromising quality of care or availability of performance data?

3. Technical Standards

VB-11. What specific interoperability challenges have you encountered in implementing value-based care programs?

VB-12. What technology standardization would preserve program-specific flexibility while promoting innovation in APM technology implementation?

VB-13. What improvements to existing criteria and standards would better support value-based care capabilities while reducing provider burden?

VB-14. How could implementing digital identity credentials improve value-based care delivery and outcomes?

VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR endpoints directory to maximize its effectiveness?